Teensy, 5V Logic

File: Teensy, 5V Logic.kicad_sch

Outputs

File: outputs.kicad_sch

Power

File: power.kicad_sch

UART

File: uart.kicad_sch

Spindle Control

File: Spindle Control_sch.kicad_sch

Isolated input

File: isolated input_sch.kicad_sch

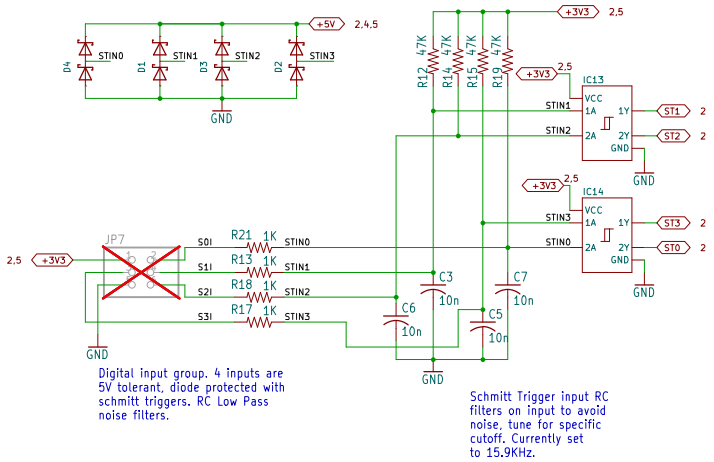
Brookwood Design

Sheet: /

File: V303.kicad_sch

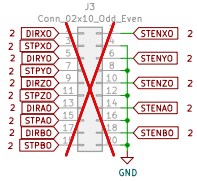
Title: T41U5XBB

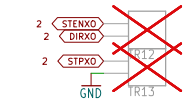
Size: User	Date: 2026-10-14	**Rev: V3.03**
KiCad E.D.A. 8.0.9		Id: 1/7



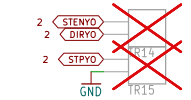
Digital input group. 4 inputs are 5V tolerant, diode protected with schmitt triggers. RC Low Pass noise filters.

Schmitt Trigger input RC filters on input to avoid noise, tune for specific cutoff. Currently set to 15.9KHz.

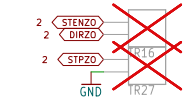




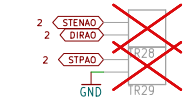
X Axis driver interface.



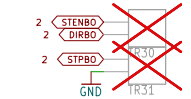
Y Axis driver interface.



Z Axis driver interface.



A Axis driver interface.

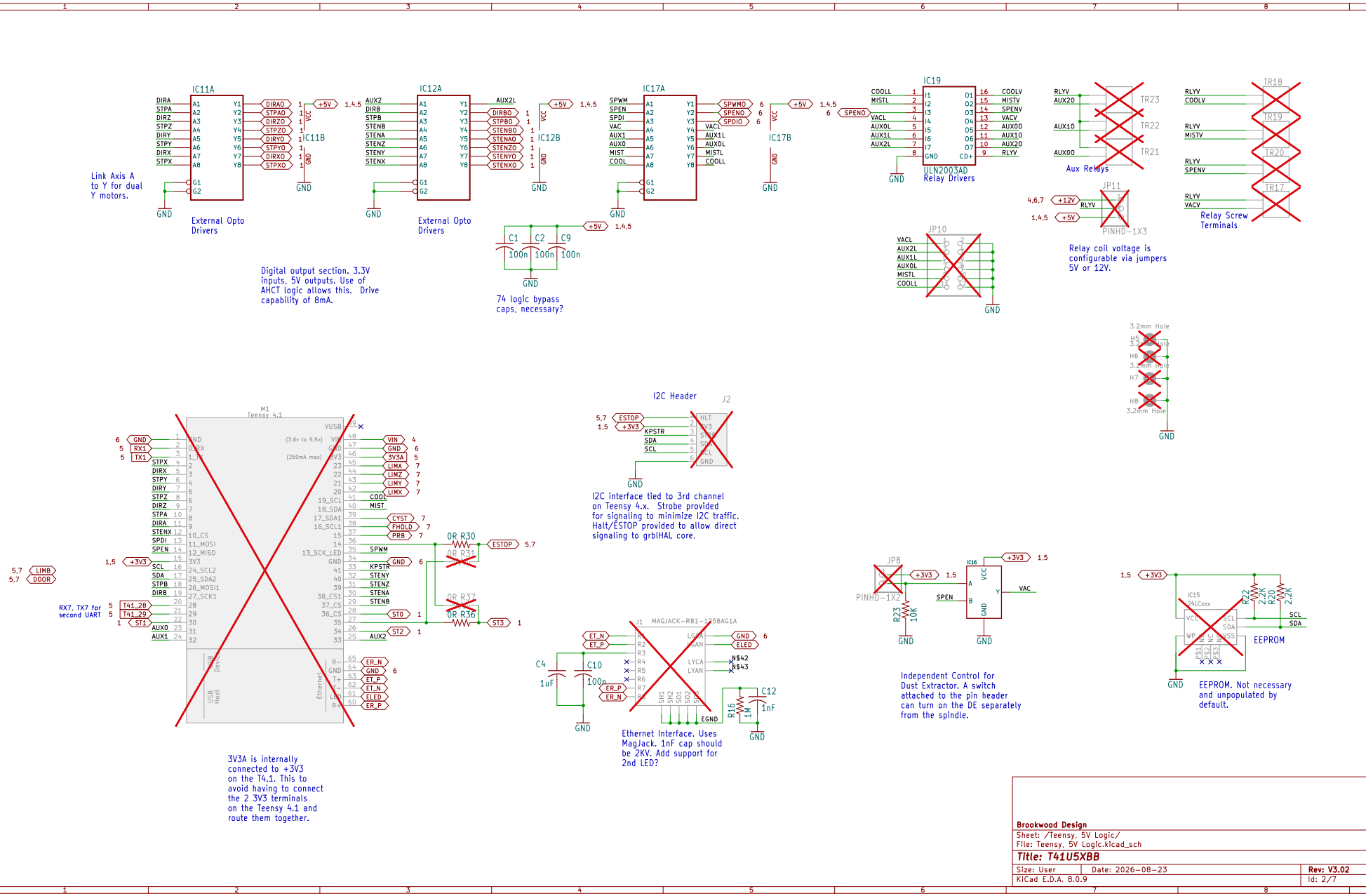


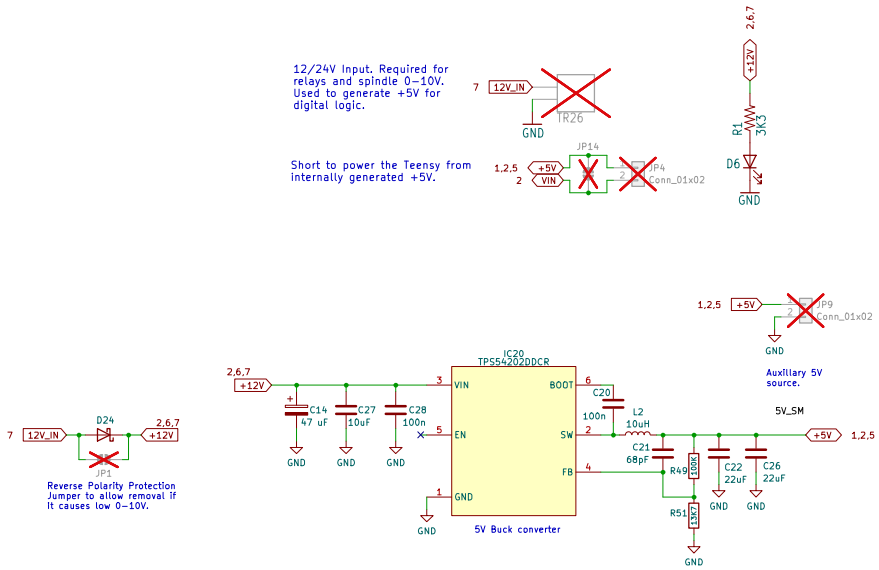
B Axis driver interface.

Brookwood Design
Sheet: /Outputs/
File: outputs.kicad_sch

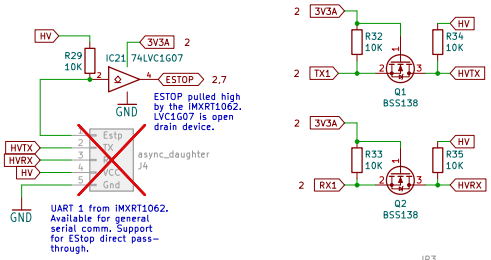
Title: T41U5XBB

Size: User	Date: 2026-08-23	Rev: V3.02
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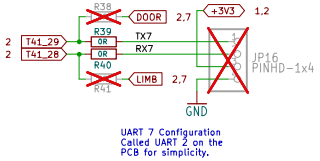

IC11A
IC11B
IC12A
IC12B
IC17A
IC17B
IC19
ULN2003AD
Relay Drivers
Link Axis A to Y for dual Y motors.
External Opto Drivers
External Opto Drivers
Digital output section. 3.3V inputs, 5V outputs. Use of AHCT logic allows this. Drive capability of 8mA.
74 logic bypass caps, necessary?
Aux Relays
Relay coil voltage is configurable via jumpers 5V or 12V.
Relay Screw Terminals
I2C Header
I2C interface tied to 3rd channel on Teensy 4.x. Strobe provided for signaling to minimize I2C traffic. Halt/ESTOP provided to allow direct signaling to grblHAL core.
RX7, TX7 for second UART
3V3A is internally connected to +3V3 on the T4.1. This to avoid having to connect the 2 3V3 terminals on the Teensy 4.1 and route them together.
Ethernet Interface. Uses MagJack. 1nF cap should be 2KV. Add support for 2nd LED?
Independent Control for Dust Extractor. A switch attached to the pin header can turn on the DE separately from the spindle.
EEPROM
EEPROM. Not necessary and unpopulated by default.
Brookwood Design
Sheet: /Teensy, 5V Logic/
File: Teensy, 5V Logic.kicad_sch
Title: T41U5XBB
Size: User
Date: 2026-08-23
Rev: V3.02
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Brookwood Design		
Sheet: /Power/		
File: power.kicad_sch		
Title: T41U5XBB		
Size: User	Date: 2026-08-23	**Rev: V3.02**
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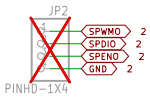
Brookwood Design

Sheet: /UART/

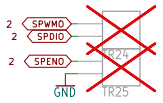
File: uart.kicad_sch

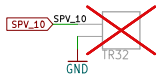
Title: T41U5XBB

Size: User	Date: 2026-08-23	Rev: V3.03
KiCad E.D.A. 8.0.9		Id: 5/7

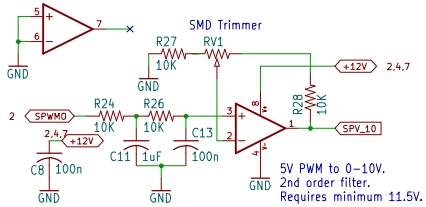


Spindle Control interfaces. PWM, Direction, Enable and 0-10V analog.



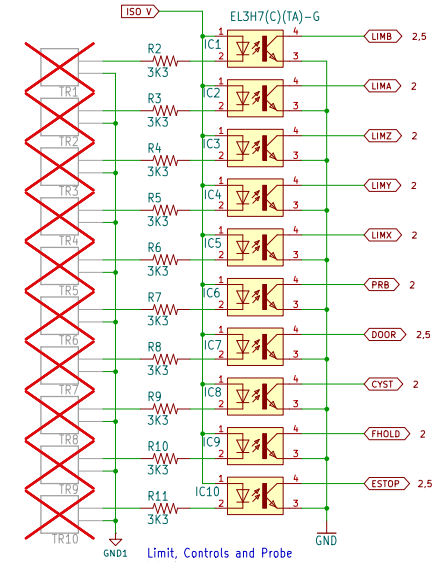


Spindle Control. Supports direct PWM and on/off (via Spindle enable) as well as 0-10V control. 10V output is adjustable via trim pot. Consider replacing LM358 with smaller opamp, single channel (JLC LM358 is Basic part and lowest cost Op Amp).



5V PWM to 0-10V.
2nd order filter.
Requires minimum 11.5V.

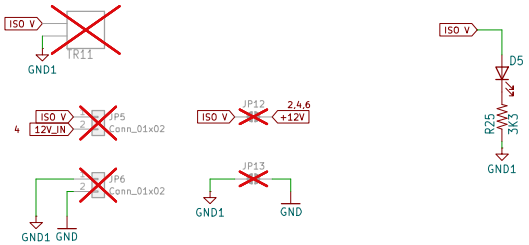
Brookwood Design		
Sheet: /Spindle Control/		
File: Spindle Control_sch.kicad_sch		
Title: T41U5XBB		
Size: User	Date: 2026-08-23	**Rev: V3.02**
KiCad E.D.A. 8.0.9		Id: 6/7



GND1 is the isolated ground plane.

Limit, Controls and Probe Interfaces.

12/24V operation on LED side. 3.3V operation on phototransistor side. LED If set to slightly higher than 10 mA.



Isolated Input side. Can be strapped to main power input or separately powered via isolated PSU

Brookwood Design

Sheet: /Isolated input/

File: isolated input_sch.kicad_sch

Title: T41U5XBB

Size: User	Date: 2026-08-23	**Rev: V3.02**
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